

Part 3 of 4

Defence Lasers and Optronic Systems: Semiconductor Diode Laser Electronics

Semiconductor diode lasers are close second to solid-state lasers in usage as far as tactical military applications of lasers are concerned. Laser aiming modules fitted on small arms and used to enhance night-fighting capability of infantry soldiers, short-to-medium range eye-safe laser rangefinders used for observation and surveillance, laser proximity sensors and short-range laser dazzlers are some prominent examples. Focus in this part of the article is on semiconductor diode laser electronics

DR ANIL K. MAINI AND NAKUL MAINI

While semiconductor diode lasers, or simply laser diodes, are extensively used in a range of laser devices intended for tactical military applications, such as short and medium-range laser rangefinders, proximity sensors, short-range laser dazzlers and laser aiming devices, it needs to be emphasised here that they are invariably used as the optical pumping source for all Nd:YAG laser-based military systems.

Laser-diode electronics primarily comprises a constant-current source that can provide to the forward-biased laser diode the desired magnitude of current and also has in-built features to provide protection to the device against all those parameters it is adversely sensitive to. The other important circuit block is the temperature controller that can maintain the laser diode junction at the desired value of constant temperature irrespective of ambient temperature. The need for a precise constant-current source and constant-temperature operation arises from the dependence of laser-diode wavelength on drive current and operating temperature. Not all laser-diode applications, including military applications, have stringent requirements of ultra-stable wavelength necessitating a high degree of current and temperature stabilisation, though for efficient and reliable operation, stability of

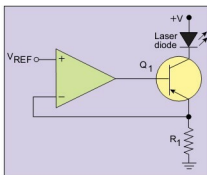


Fig. 1: Constant-current mode

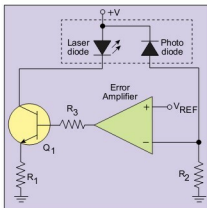


Fig. 2: Constant-output-power mode

drive current and operating temperature are always desirable.

Although laser diodes exhibit excellent reliability under ideal operating conditions, these are highly susceptible to damage due to excessive drive current, electrostatic discharge (ESD) and transients. Laser diode damage manifests itself in the form of one or more of the following prob-

lems, namely, reduced output power, shift in threshold current and increase in beam divergence and failure to laser action, thus producing an LED-like output only.

Relevant damage mechanisms are overcurrent, overheating, current spikes and power-line surges. Improper grounding and shielding lead to radiated electrical transients. Improper power-line conditioning also leads to severe fast transients. ESD caused by improper handling during storage, transport and mounting is considered as the single leading cause of premature laser diode failure.

Constant-current and constant-power operational modes

There can be two possible circuit topologies the laser diodes can be operated in, namely, constant-current and constant-output power modes. Operating temperature is of course an important factor in both the modes. In view of dependence of laser diodes' performance on operating temperature and drive current, the preferred mode of their operation is the constant drive current with precise control of operating temperature. Constant-current operation without temperature control is generally not desirable. Even at constant drive current, output power would increase with decrease in operating temperature. With significant